From Mathematics to Generic Programming

Companion repository: corrected solution

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Exercise 4.4: Odd and even squares

Proposition. For any odd square x, there is an even square y such that $x + y$ is a square.

Proof. Start with $9 + 16 = 25$, or $3^2 + 4^2 = 5^2$. We need an even *square*, not merely an even number. Write $x = (2k + 1)^2 = 4k(k + 1) + 1$. Since one of the consecutive integers k and $k + 1$ is even, $x \equiv 1 \pmod 8$. Consequently $m = (x - 1)/2$ is even (indeed, a multiple of four). Set

$$y = m^2 = \left(\frac{x - 1}{2}\right)^2.$$

Then y is an even square and, since $x = 2m + 1$,

$$x + y = 2m + 1 + m^2 = (m + 1)^2.$$

Boundary case. For $x = 1$, this construction gives $m = 0$ and $y = 0 = 0^2$, so $1 + 0 = 1^2$. The exercise does not explicitly say "positive squares." Under the convention that squares include 0^2, the proposition includes this case. If squares are required to be positive, the proposition must instead assume $x > 1$: two positive integer squares cannot differ by one, so no positive square y works when $x = 1$.